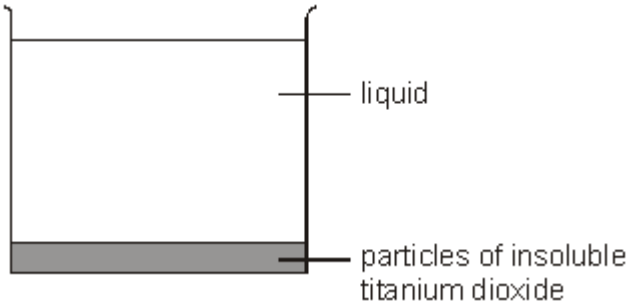
Q1. (a) Samantha opened a tin of white paint. The paint consisted of a liquid and particles of titanium dioxide that are insoluble in the liquid.
The paint had separated into two layers, as shown below.



(i) What type of substance is the paint?
Tick the correct box.

a compound ☐ an element ☐ a mixture ☐

1 mark

(ii) What type of substance is titanium dioxide?
Tick the correct box.

a compound ☐ an element ☐ a mixture ☐

1 mark

(iii) Why did the particles of insoluble titanium dioxide sink to the bottom?

.....

.....

1 mark

(b) Samantha stirred the paint and used it to paint a window frame.
She got some of the paint on the glass.



Samantha could **not** get the paint off the glass with water.
When she used a different liquid called white spirit the paint came off.

Why could she remove the paint with white spirit but **not** with water?

.....

.....

1 mark
maximum 4 marks

Mark schemes

Q1.

(a) (i) a mixture ✓

if more than one box is ticked, award no mark

1 (L5)

(ii) a compound ✓

if more than one box is ticked, award no mark

1 (L5)

(iii) any **one** from

- they are denser than the liquid

  *accept 'it is heavier than the liquid **or** the paint'*

  *accept 'the solid particles are more dense **or**heavier **or** too heavy'*
  accept 'the solid is denser'

  *do **not** accept 'solid particles are heavy' without a comparison*
  ***or** qualifier eg 'too heavy'*

- the liquid is less dense than the solid

  *accept 'the liquid is less dense' **or** 'the liquid is lighter'*

1 (L5)

(b) any **one** from

- it is insoluble in water
- water is not a solvent for the paint
- it dissolves in white spirit
- white spirit is a solvent for the paint

  'it is waterproof' is insufficient

1 (L5)

[4]